

Understanding the Context for Data Presentations Transcript

Now, we will understand a new concept called context, how context helps somebody to design better visuals. In fact, context is the single most important feature that one has to remember while designing any data visualisation. It is not something that is objective and readily observable like a chart, it is an approach, it is something that helps you understand to whom you are talking to, it is the reason that you are doing a data visualisation, all these would form part of context.

Context is very important for two reasons, it helps as a directional post in the myriad of data that we are trying to understand. In today's world, you are getting large volumes of data with different variables and most often the point is to collect appropriate information. And how will you choose this information is guided by the context and what forms the parts of this context.

The central piece of context is the audience to whom you are going to cater, once you anchor your data visualisation on the basis of the audience to whom you are catering, you are more sustainably to give a better output. Always ask this question to whom I am conveying this information, second thing what are they looking from this information and do I have the sufficient data with me to give out this information. Once you have these 3 or 4 questions ready, you have a broad understanding of the context and as you go into the process, the context becomes more clearer and it helps you build more visuals or it helps you build visuals with more clarity.

This is something that you should keep always in the back of your mind while designing visuals and trying to tell a story. I am again emphasising context is a very important process for developing better visualisations, you have to spend sufficient time on building your context. The more you try to get a better context, the greater will be the clarity and efficiency of your data visual and you have to always ask the question what is the intended purpose of your data visualisation, who are the audience and these questions help you better build a better context.

Once you have the context, you can then navigate through building the data visuals with a clear sense of purpose and therefore, tell a better story. Whenever we come about talking about context, we have to tailor them to very specific needs, you have a very broad understanding of the context, but as you go through your exploratory process and explanatory process, your context also gets refined. And once you reach a stage where you have clearly identified the storyline and the points that you want to convey, the context and the purpose should be clearly matching.

Once you do have this in your place, you can then start building the data visualisation and story and it will be effectively communicating what you wanted to tell. Let me give you an example. We have looked at vote or turn out ratio.

Now, I will try to bring the role of context in this and try to explain how context becomes very important. For example, you have taken vote or turn out ratio and we have come

seen that some states are doing better, some states are doing below national average. The question here is to whom you are catering this information to.

Let us say you are a national newspaper right and you are trying to portray the on the day after elections what has happened in the elections. And for you the simple answer is try to showcase the slide with the graphic that is on the your leftmost side. You are a national newspaper, your focus is on delivering the news that is at the national level.

So, you try to portray a single graphic with all the states data and try to show the highest state with the highest vote or turn out and the state with the lowest turn out, national average and you might do a comparison with 2019 and bring out the trends and also show the increase in the electoral base so on and so forth. Here the choice of visual, the communication, the explanatory part everything is driven by the purpose that is to showcase a national average picture because you are a national daily or a national newspaper. Now let us shift the context, the audience is different now.

The audience let us say is a regional newspaper or a party a national party looking at the regional performance of states. Now the second visual would be more appropriate accordingly the explanatory and explanatory parts also get modified, you would be rather looking at what is the state level performance and mind you in our original data set we did not have the state level variable, the region in which the state belongs to we had to incorporate that and get this visual done. Again here the context is driven by the audience or the purpose with which we are trying to answer certain questions.

Here it is a state level ah the regional party or a national party looking at the regional performance or a regional newspaper looking at the state wise performance to see where the state vis-a-vis its regional neighbours or partners is performing. So the second visual becomes important. Now let us talk about the third visual from a different context.

You are neither a party nor a newspaper but somebody interested in improving the voter turnout let us say it is election commission of India. Election commission of India is looking at reasons to improve voter turnout in this case the state level voter turnout per se does not matter. What is more interesting for election commission of India is the per booth average number of voters whether it is affecting the voter turnout and we found a slight negative relationship saying that if you have higher number of people casting their votes in a particular booth voter turnout is slightly to go down and therefore it makes the case for increasing the number of booths.

However it is not cost ah it is not cost free. From election commission of India's perspective it is trying to look at reasons for improving voter turnout. In this case the third visual would become more appropriate where we have seen average electors per booth and its relation to voter turnout ratio and we found a slight negative relationship and in this relationship it can be said or at least said as a reason that if you have higher number of voters per booth the voter turnout is likely to be lower and this can provide insight to election commission to increase the number of booths in a

particular state or a particular region to improve the voter turnout.

However again as I would emphasise in this case the choice of visual the exploratory analysis than the explanatory analysis are driven by the person who is trying to look at data or the person who is expected to look at the data in this case it is election commission of India. Point that I am trying to drive home is these three visuals come from the same data but go through different paths to become a final visual and this is driven by the context. So, one has to understand and appreciate the role that context plays in delivering effective visuals.

I can convey the same information for the national level news channel using a scatter plot saying that average voter turnout is 66 percent I can colour the same scatter plot in the same scatter plot the regions that are above the trend line as green and below the trend line as red. It conveys technically speaking it conveys same information the ones which are above trend line have higher than national average voting and the ones below have ah less than national average trending or I can colour them differently green same green colour as ah brick. Let me emphasise the role of context with one more example here I can as well use the scatter plot for a national ah news channel and how that would be wrong or inefficient.

It also conveys a similar information it says the national average voting is 66 percent and the states which are higher the national average can be coloured in green and the ones which are colour less than national average can be coloured in red. So, imagine a situation where you have a scatter plot where the ones which are above the ah national average are coloured in green and the ones which are below are coloured in red. What is happening? It is conveying the same information as a horizontal bar chart, but it is cumbersome it does not readily convey the information to the audience which are general people who might not understand a scatter plot rather than horizontal bar speaks very well to them as compared to scatter plot.

But for election commission of India horizontal bar is useless scatter plot is more useful in trying to drive the reason. So, it is very important to build the context first before we deliver any visual or tell a story out of it. Now, let us see what kind of questions can help you to set the context first and foremost you have to ask the background of the audience to whom you are trying to convey this information.

Then we have to ask whether the audience is whether there is an audience or it is a decision maker accordingly also the question the presentation of the visual changes and whether the audience have any biases or there are any sudden friction points that can come which we can address. And however, finally, and the most importantly ah how much space you have to present your data it it need not be in terms of physical space, but also in terms of time or that the audience can devote to you or the decision maker can devote to you. You also have to do some understanding on that and derive your context if you had all the space you can provide 4, 5 visuals, but if you had only limited space or limited time then that also becomes part of context to give most



effective visual for that particular time and space.

So, all these forms part of developing questions all these questions form part of developing the context which will help you to determine ah which will help you to design better visualisations . So, we will now be taking a quiz to understand your concepts around the context .

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